Galaxy Morphology Classification Under Observational Domain Shift: A Cross-Model Study on GalaxyZoo2

Danyal Ahmed
Khoury College of Computer Sciences
Northeastern University
dahmed@northeastern.edu

Abstract*—Automated galaxy morphology classifiers are routinely trained on confidently-labeled galaxies at low redshift, then applied across surveys spanning observational regimes that the training distribution did not cover. I study how three model families of increasing capacity—a linear SVM and a gradient-boosted tree ensemble on engineered features, and a compact CNN trained end-to-end—degrade under a redshift-defined domain shift on GalaxyZoo2. Training was carried out on confident labels at $z < 0.102$ and evaluating on $z \geq 0.102$. The CNN achieves the highest macro-F1 in both regimes (0.77 IID, 0.67 OOD) and the smallest aggregate degradation (-0.11). Beyond aggregate numbers, three primary patterns emerge: (1) at matched redshift, domain membership alone costs the SVM 7.5 macro-F1 points versus 3.2 for the CNN, isolating a domain effect from the redshift effect; (2) all three models collapse on elliptical galaxies at high redshift, with per-class accuracy dropping $\sim$50 points from low to high-z, while "accuracy" on face-on non-spirals rises monotonically; (3) the face-on non-spiral "accuracy" gain is an over-prediction artifact, indicated by a rise in recall from 0.64 to 0.92 while precision falls from 0.28 to 0.23 across the redshift tail, reflecting misclassified ellipticals being absorbed into the non-spiral class. Per-unit of redshift, macro-F1 slopes are -2.37 (SVM), -3.49 (GBDT), -2.51 (CNN), showing the GBDT to be uniquely brittle despite being higher-capacity than SVM. The substantial failure to classify elliptical galaxies is shared across all three models and reflects observational information loss rather than modeling choices.

I. Introduction

Galaxy morphology classification is a long-standing astronomical task, as understanding the distribution of galaxy shapes and structure informs cosmological models, current theories on galactic evolution, and research on large-scale galactic structures. Modern surveys like the Sloan Digital Sky Survey (SDSS) image hundreds of millions of galaxies, making automated classification a necessity for gathering information on the cosmos.

Most automated classifiers are evaluated on held-out galaxies drawn from the same distribution as their training data, typically low-redshift galaxies with high-quality images and confident human labels. In practice, however, the same classifier is applied to galaxies spanning a range of observational conditions. At higher redshift, corresponding to greater distance from the observer (us), galaxies are smaller in angular size, fainter, and more susceptible to occlusion, smearing, and sky-dominated noise. Features that cleanly distinguish morphological classes at $z \approx 0.05$, such as spiral arms or bars, may become ambiguous or invisible at $z > 0.15$, even when the underlying galaxy is intrinsically similar. This is a textbook example of distribution shift: the input distribution changes in a systematic, measurable way across the deployment domain, while label semantics remain fixed.

I study how three classifiers of increasing representational capacity degrade under a redshift-defined domain shift on the GalaxyZoo2 (GZ2) [1] dataset. Using Hart et al.'s debiased vote fractions [2] to construct a four-class morphological taxonomy, I train a linear SVM, a gradient-boosted tree ensemble, and a compact CNN on galaxies at $z < 0.102$ (the 70th percentile of the labeled population), then evaluate on the held-out $z \geq 0.102$ tail. Beyond reporting aggregate macro-F1 degradation, I present three analyses that strengthen the cross-model comparison: a controlled bin-matched comparison that isolates domain membership from redshift effects, per-unit-redshift degradation slopes that quantify sensitivity, and a precision-vs-recall decomposition that exposes a cross-model over-prediction artifact on the face-on non-spiral class.

The principal findings are that the CNN achieves the highest performance and smallest degradation across all measurements, but that a large component of the observed degradation reflects a shared failure mode (a dramatic elliptical collapse at high redshift) that is not remedied by model capacity. The GBDT is uniquely brittle despite being higher-capacity than the SVM, which I attribute to the mismatch between its hard decision thresholds and the smooth shift in feature distributions at the domain boundary.

II. Methods

A. Dataset

The main GalaxyZoo2 [1] image corpus comprises roughly 240k 424×424 RGB JPEGs. I merged three CSVs: `gz2_hart16.csv` (debiased vote fractions), `gz2sample.csv` (redshift and photometric metadata), and `gz2_filename_mapping.csv` (map of object IDs to image assets). The vote fraction data used is that released by Hart et al. [2], which corrected redshift-dependent bias on vote fractions across the decision tree.

Class separation collapsed the 37-node GZ2 decision tree to four terminal classes: Elliptical, Edge-on disk, Face-on spiral, Face-on non-spiral. This set of classes is chosen because they are defined by the presence of detectable features rather than their absence, producing a more coherent vote-fraction label than taxonomies relying on the absence of signal. Hard labels were constructed using a confidence threshold of 0.7 on each relevant decision-tree branch. Rows that did not meet the threshold were discarded as overly ambiguous.

1) Domain Split: The central experimental design is a train/test split along a redshift boundary. After labeling, I partitioned the confident-label set at the 70th percentile of redshift, yielding:

- **Easy** (training domain): $z < 0.102$, 124,798 galaxies
- **Hard** (OOD test set): $z \geq 0.102$, 53,486 galaxies

The easy set was further stratified-subsampled to 100,000 and 70/15/15 split into train/val/test for IID-regime evaluation. The hard set was used in its entirety for OOD evaluation. All hyperparameters were tuned on the easy-domain validation split only, with no look-ahead to OOD data.

Two properties of this split are worth noting. First, both the IID and OOD sets are filtered to confidently-labeled galaxies, which likely understates domain shift severity: at genuine high redshift, labels themselves become noisier because human annotators struggle with the same observational degradations the models face. Second, the OOD set has a different class mix than the IID set (notably more face-on spirals and fewer confident ellipticals as a fraction), which confounds raw accuracy but is partly corrected for by reporting macro-F1.

2) Image Preprocessing and Feature Extraction: Images were center-cropped from 424×424 to 224×224. For the CNN, per-channel RGB mean and standard deviation were computed from a 10,000-image subset of the training split and applied to all inputs:

$$\mu \approx [0.046, 0.041, 0.031], \quad \sigma \approx [0.090, 0.076, 0.067]$$

For the classical models, each image was converted to luminance and summarized by a 207-dimensional feature vector $\phi(x)$ comprising: concentration index, asymmetry, smoothness, Gini coefficient, M_{20}, ellipticity, position-angle orientation, normalized Petrosian radius (8 scalars); a 10-bin radial flux profile; and a three-level spatial pyramid gradient histogram with 9 orientation bins (189 values). Features were standardized using training-split statistics.

B. Models

1) Linear SVM: A one-vs-rest linear SVM (`sklearn.svm.LinearSVC`) with `class_weight="balanced"`. Grid search was conducted over $C \in \{0.01, 0.1, 1.0, 10.0\}$ and the best value selected by validation macro-F1.

2) Gradient-Boosted Trees: A histogram-based gradient-boosted ensemble with balanced class weights, early stopping on a 10% internal validation subset with a patience of 20 intervals. Grid search was conducted over learning rate $\in \{0.05, 0.10\}$ and max iterations $\in \{200, 400\}$, with the best combination selected by validation macro-F1.

For both classical models, the final model was refit on the combined train+val split before evaluation on test and OOD splits.

3) CNN: The CNN is a compact from-scratch architecture consisting of a 7×7 stride-2 stem (32 channels), followed by four blocks of increasing channel depth (64, 128, 192, 256), each containing one stride-1 and one stride-2 convolution layer. Output feeds into global average pooling, a dropout layer ($p = 0.4$), and a linear layer to 4 output logits. Each convolution layer is followed by BatchNorm with ReLU activation. Total parameter count: $\sim$1.87M.

Training used the AdamW optimizer with initial learning rate 3×10^{-4}, weight decay 1×10^{-4}, and cosine annealing to zero over 30 epochs. Batch size was fixed at 128. Training-set images were augmented with random rotations up to $180°$ followed by independent horizontal and vertical flips ($p = 0.5$ each), applied before center cropping to trim rotation artifacts. A `WeightedRandomSampler` with inverse-frequency for each set of sample weights was used to handle class imbalance. The loss was standard hard-label cross-entropy. The evaluation checkpoint was the epoch with lowest validation loss (epoch 20 of 30). Training used a single NVIDIA RTX 5070Ti.

C. Evaluation

The primary metrics are top-1 accuracy, macro-averaged F1, per-class accuracy, and the 4×4 confusion matrix, computed on both IID test (15,000 galaxies) and OOD hard set (53,486 galaxies). To characterize the transition from low to high redshift, I partitioned the union of IID and OOD galaxies into 8 quantile-based redshift bins (roughly 8,560 galaxies each) and computed per-bin macro-F1 and per-class accuracy. Bootstrap 95% confidence intervals on macro-F1 used 200 iterations per bin, with replacement.

To separate the contribution of pure domain membership from the redshift-induced drop, I extracted the single bin in which both IID (6,439) and OOD (2,120) galaxies appear with substantial counts (bin 1 at median $z \approx 0.089$) and directly compared IID and OOD metrics within it.

To quantify rates of degradation, I fit linear regressions of macro-F1 and per-class accuracy against bin-center redshift over the OOD-only bins (bins 2-7).

To separate the face-on non-spiral "accuracy" rise from a genuine improvement, I computed per-class precision, recall, and F1 per bin.

III. Results

A. Overall Performance

Table I reports macro-F1 and accuracy for each model on each domain, with the IID $\rightarrow$ OOD degradation. By macro-F1, the CNN outperforms both classical models in both regimes and shows the smallest degradation (-0.107). The GBDT exceeds the SVM on IID (0.633 vs 0.598) but degrades more sharply, closing the gap on OOD (0.515 vs 0.452). Accuracy tells a different story: the CNN actually has the largest accuracy degradation (-0.223), which reflects the shift in class proportions between IID and OOD, a factor macro-F1 partially controls for but accuracy does not.

B. Performance vs. Redshift

Figure 1 shows macro-F1 against median redshift for each bin. Within IID (bins 0-1), models order CNN $>$ GBDT $>$ SVM at a spread of roughly 20 points. Across the OOD tail (bins 2-7), the CNN retains the top position but its absolute advantage shrinks, and the SVM overtakes the GBDT in the highest-redshift bin (bin 7, $z = 0.176$): SVM macro-F1 $= 0.338$, GBDT $= 0.311$. The CNN reaches 0.507 in the same bin, a 16-point lead that is comparable in absolute size to its IID lead but a markedly larger relative share of the remaining performance.

Linear regression of macro-F1 on redshift across OOD bins yields slopes of -2.37 (SVM), -3.49 (GBDT), and -2.51 (CNN) per unit z. The CNN's slope is statistically indistinguishable from the SVM's despite an $\sim$0.22 difference in absolute performance, which means the CNN's superior OOD macro-F1 is a consistent vertical offset rather than a slower rate of degradation. The GBDT degrades roughly 40% faster per unit z than either alternative.

C. Isolating Domain Membership from Redshift

Bin 1 (median $z = 0.089$) contains 6,439 IID galaxies and 2,120 OOD galaxies at essentially the same redshift. Any performance gap between them cannot be attributed to redshift, and therefore reflect other properties of the domain partition (photometric conditions, galaxy selection, labeling characteristics that correlate with the IID/OOD assignment but not with z). Table II reports this controlled comparison.

At identical redshift, the CNN loses less than half the macro-F1 the SVM loses when crossing the domain boundary. This demonstrates that the CNN's advantage is persistent when redshift is controlled for. The classical models also show a substantive controlled-domain penalty, confirming that pure domain membership contributes measurable degradation distinct from redshift.

D. Per-Class Analysis

Table III reports per-class accuracy on IID and OOD, and the IID $\rightarrow$ OOD change. Ellipticals collapse across all three models, edge-on disks are near-uniformly domain-robust across all three models (drops of 4-6 points), and face-on spirals drop 14-16 points uniformly, suggesting the loss of arm visibility at high redshift is a physical limitation no model can overcome. Face-on non-spirals "improve" by 9-25 points, an artifact addressed in Section III-E.

Figure 2 shows per-class accuracy against redshift for all three models, with the vertical dotted line marking the $z = 0.102$ domain boundary. The elliptical subplot is the most visually striking: all three models drop along nearly parallel lines from near-unity accuracy (in the CNN's case) to well below 0.30 in the highest-redshift bin. Per-unit-redshift slopes for ellipticals are -5.16 (SVM), -6.46 (GBDT), and -6.00 (CNN), 2-3$\times$ steeper than the overall macro-F1 slopes.

TABLE I: Overall performance by model and domain

Model	Macro-F1			Accuracy		
	IID	OOD	Δ	IID	OOD	Δ
Linear SVM	0.598	0.452	−0.146	0.734	0.576	−0.158
GBDT	0.633	0.515	−0.118	0.718	0.518	−0.200
CNN	**0.772**	**0.665**	**−0.107**	**0.859**	**0.636**	−0.223

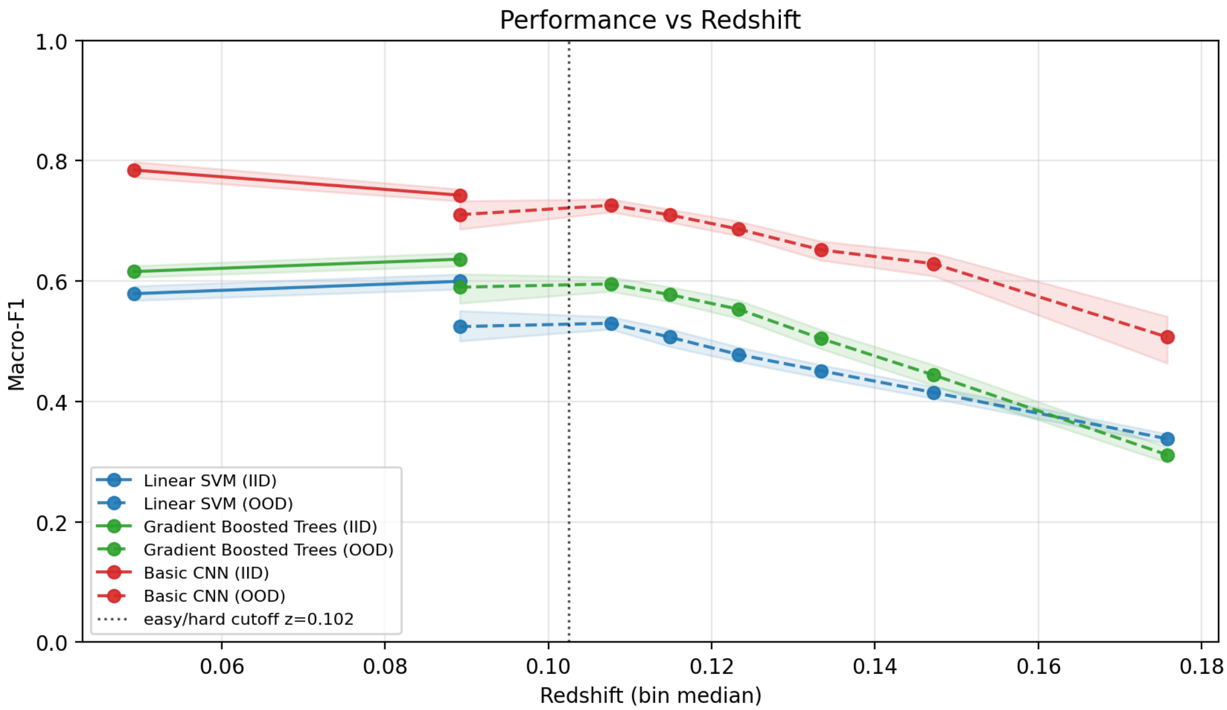

Fig. 1: Macro-F1 vs. redshift bin. Solid lines show IID-designated galaxies (bins 0-1); dashed lines show OOD-designated galaxies (bins 1-7, with overlap at bin 1).

TABLE II: Controlled comparison at bin 1 ($z \approx 0.089$)

Model	IID macro-F1	OOD macro-F1	Δ
Linear SVM	0.600	0.524	−0.075
GBDT	0.636	0.590	−0.047
CNN	0.743	0.710	**−0.032**

E. Face-on Non-Spiral Over-prediction

The face-on non-spiral class shows an consistent rise in accuracy as redshift increases, counterintuitive for a domain-shift into a harder regime. Decomposing per-class accuracy into precision and recall reveals the source (Table IV).

The per-class accuracy is actually recall—fraction of true non-spirals correctly labeled as non-spiral. The rising recall is real in that sense. However, precision (fraction of non-spiral predictions that are actually non-spirals) falls in tandem, and F1 is flat or slightly declining across the redshift tail. The mechanism is that as redshift increases, models progressively mislabel ellipticals (and to a lesser extent spirals) as non-spirals—non-spiral becomes a "dumping ground" for objects that have lost their distinguishing features. This is the same phenomenon as the elliptical collapse, viewed from the other side: class-0 loses recall, class-3 gains false positives.

The SVM pattern is qualitatively different: both precision and recall remain low, reflecting the SVM's general reluctance to predict non-spiral at any redshift (IID non-spiral accuracy was already only 0.14, Table III).

TABLE III: Per-class accuracy (IID / OOD / Δ)

	SVM			GBDT			CNN		
Class	IID	OOD	Δ	IID	OOD	Δ	IID	OOD	Δ
Elliptical	.742	.644	$-.10$	.687	.388	$-.30$	.901	.511	$-.39$
Edge-on disk	.838	.795	$-.04$	.905	.848	$-.06$	.946	.893	$-.05$
Face-on spiral	.770	.609	$-.16$	.737	.573	$-.16$	.835	.687	$-.15$
Face-on non-spir.	.140	.232	$+.09$	.435	.687	$+.25$	.589	.818	$+.23$

TABLE IV: Face-on non-spiral precision and recall across OOD bins

		CNN			GBDT		
Bin	z	P	R	F1	P	R	F1
2	.108	.33	.76	.45	.24	.58	.34
3	.115	.30	.77	.44	.23	.63	.34
4	.123	.30	.79	.44	.23	.65	.34
5	.133	.28	.82	.41	.22	.72	.34
6	.147	.22	.86	.35	.17	.73	.27
7	.176	.23	.92	.37	.20	.81	.31

F. *Confusion Matrices*

Figure 3 shows confusion matrices for the CNN under both domains (IID left, OOD right). On IID, the dominant off-diagonal entries are elliptical↔non-spiral and spiral↔non-spiral confusions, each around 5-10% of the respective true class. On OOD, the elliptical→non-spiral rate jumps to roughly 32%, absorbing thousands of true ellipticals into the non-spiral predictions column. This seems to be a strong source of both the elliptical recall collapse and the non-spiral precision drop.

IV. Discussion

A. *A Shared Failure Mode*

All three models, which span a capacity range from 200 hand-crafted features plus a linear boundary, an ensemble of decision trees, and nearly two million learned CNN parameters, fail on ellipticals at high redshift. The per-unit-z slopes on elliptical accuracy (-5.16 to -6.46) are markedly steeper than the overall macro-F1 slopes and steeper than any per-class slope other than class 3, which seems to be an artifact of the same phenomenon.

I do not believe this reflects a common modeling weakness. Rather, it reflects a physical limitation of the input data. At higher redshift, small-angular-size galaxies become PSF-limited. A face-on spiral with faint arms loses them to smearing and noise, presenting as a centrally-concentrated featureless blob. An intrinsically lenticular galaxy and a highly-smeared spiral can produce visually indistinguishable images. Any classifier that trained on $z < 0.102$ galaxies has learned decision boundaries calibrated to the characteristic surface-brightness profiles at those redshifts, and those boundaries are systematically in the wrong place for profiles at $z = 0.17$. No amount of learned-feature richness compensates for information that is genuinely absent from the image.

The fact that the GBDT and CNN "improve" on non-spirals while SVM does not is consistent with this story: the SVM's linear boundary is geometrically constrained to classify the bulk of the class-3 region conservatively, so it cannot absorb mislabeled ellipticals the way tree ensembles and CNNs can. The GBDT and CNN have the representational capacity to carve a larger non-spiral region and they do so aggressively at high redshift because the features that formerly distinguished ellipticals from non-spirals are no longer there.

B. *Capacity Does Not Map to Robustness*

The GBDT is the most brittle model by every metric: steepest macro-F1 slope, largest elliptical accuracy drop, overtaken by the SVM at the extreme-z bin. This runs against the intuition that expressive models are uniformly more robust. The GBDT has substantially more effective capacity than the SVM yet is less domain-robust.

I attribute this to a mismatch between hard decision thresholds and smoothly-shifting feature distributions. A gradient-boosted tree ensemble partitions feature space along axis-aligned thresholds, thresholding individual features (concentration, Gini, etc). When feature distributions shift under domain change (e.g., all galaxies appear more concentrated because PSF smearing makes them look more centralized), the trained thresholds c^*, g^* are placed in distribution regions the trained trees never encountered. Every tree whose first split depends on a shifted feature is degraded, and the ensemble is degraded by compounding error.

The SVM's linear boundary, by contrast, represents a single hyperplane whose margin at any point degrades smoothly as the feature distribution drifts—no single threshold catastrophically misaligns. The CNN's learned features are distributed across millions of parameters,

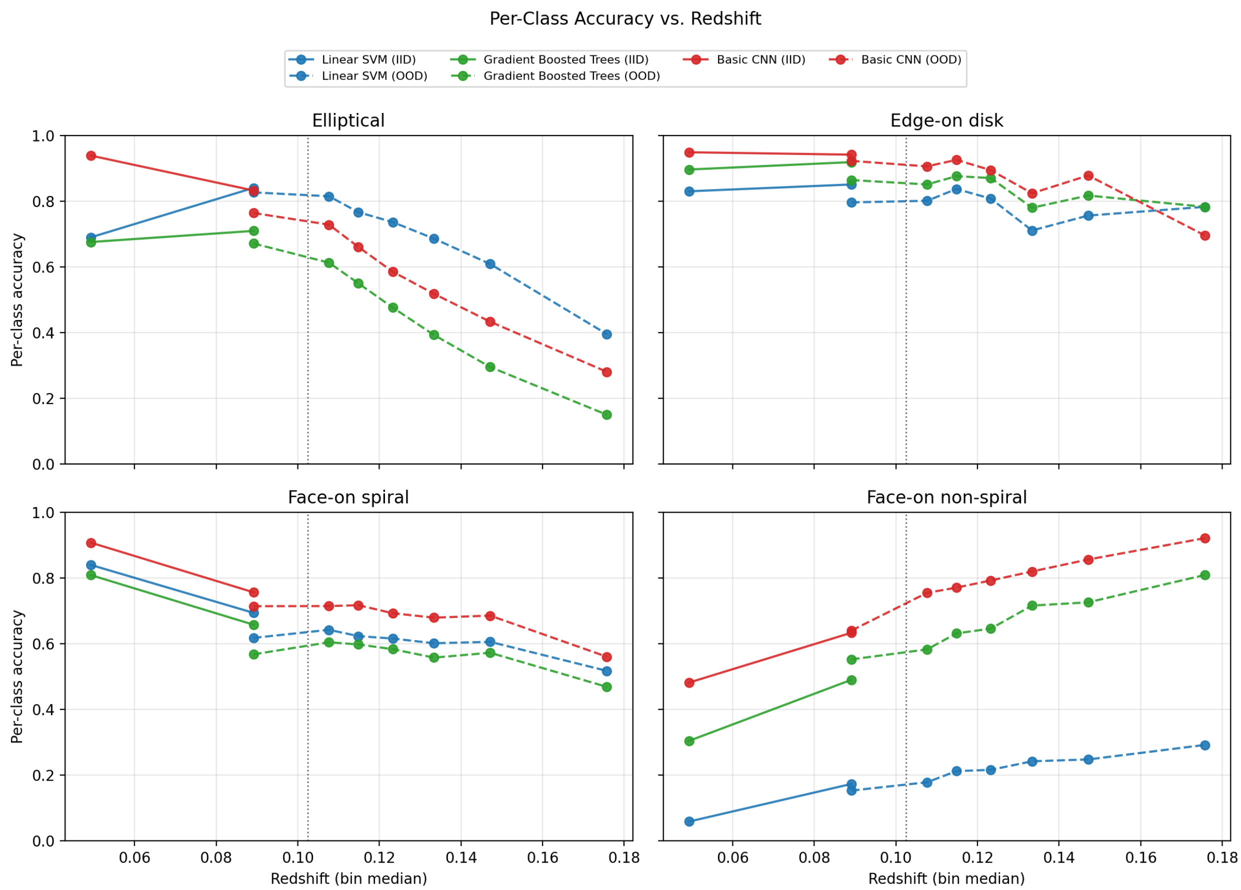

Fig. 2: Per-class accuracy vs. redshift. All three models share the failure pattern in classifying elliptical galaxies at high redshift and the monotonic rise in face-on non-spiral accuracy, both aspects of the same over-prediction artifact (see Section III-E).

making it less reliant on any individual feature being in its trained range. Both mechanisms produce more graceful degradation than the GBDT's axis-aligned thresholds.

C. Macro-F1 vs. Accuracy Diverge on OOD

In-domain, macro-F1 and accuracy order the models the same way (CNN $>$ GBDT $>$ SVM). On OOD, the macro-F1 ranking (CNN $>$ GBDT $>$ SVM) inverts the accuracy-degradation ranking (SVM smallest, CNN largest degradation). This is noteworthy, as choosing one metric over the other changes which model looks best under domain shift.

The divergence is driven by class-proportion shift: the OOD set has a much higher face-on spiral fraction and lower confident-elliptical fraction than the training distribution, so a model that happens to perform well on spirals and poorly on ellipticals looks artificially good under accuracy regardless of genuine robustness. Macro-F1, by treating all classes equally, is less sensitive to this compositional drift.

D. Limitations

Several potential confounds limit the strength of these findings.

First, both the IID and OOD evaluation sets are filtered to confidently-labeled galaxies ($\max$ vote fraction ≥ 0.7 on the relevant branches). This probably understates real-world degradation: at genuine high redshift, labels themselves become less reliable because human annotators face the same observational degradations. A more realistic OOD evaluation would include ambiguous high-z galaxies, using argmax-of-soft-target as weak ground truth.

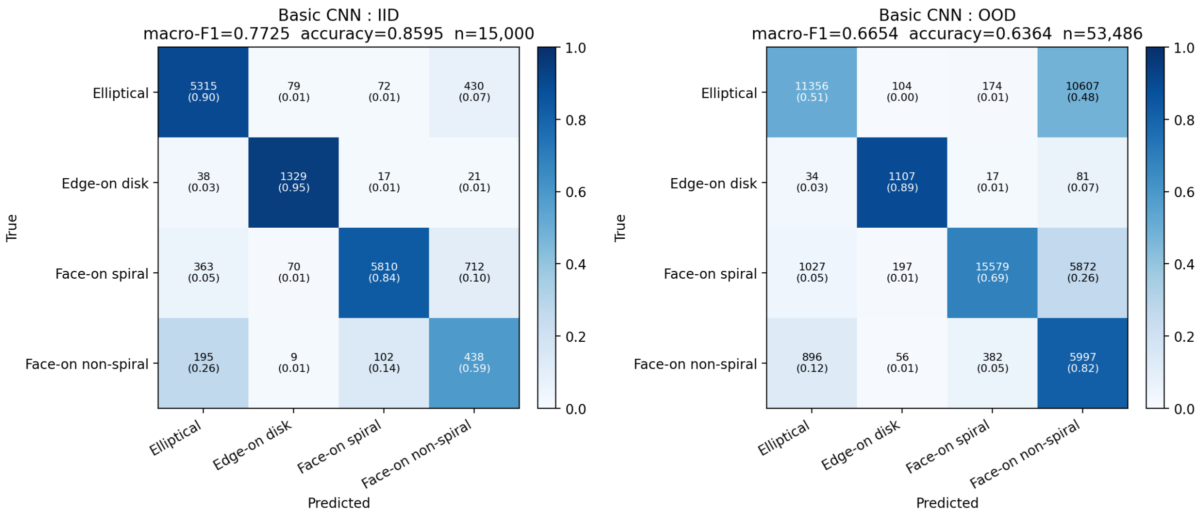

Fig. 3: CNN confusion matrices on IID (left) and OOD (right). Elliptical accuracy drops from 90% to 51%, with most lost mass landing in the face-on non-spiral column.

Second, the redshift boundary at $z = 0.102$ is the 70th percentile of the labeled population, which is a data-driven choice rather than a physically motivated transition point. A more astronomically principled split might use a seeing-limited magnitude cutoff or a specific SDSS image-quality metric. The qualitative conclusions—shared elliptical collapse, ordering of controlled-domain deltas, cross-model precision/recall divergence on non-spirals—are unlikely to depend strongly on the exact cutoff, but the quantitative slopes and percentile ranges are sensitive to it.

Third, all models were trained with a single fixed seed. Multi-seed training would establish which findings are robust to initialization and which reflect single-run sampling noise. Bootstrap CIs on macro-F1 per bin address per-bin sampling noise but not training-level variance.

Fourth, the GBDT-vs-SVM crossover in bin 7 rests on CIs that just barely separate (SVM $[0.330, 0.346]$ vs. GBDT $[0.299, 0.324]$). The difference is real but small, and the slope analysis is the stronger quantitative claim for GBDT brittleness.

V. Conclusion

This study examined how three classifier families of increasing capacity—Linear SVM, GBDT, and CNN—degrade under a redshift-defined observational domain shift on GalaxyZoo2. The CNN is the most accurate and most domain-robust by macro-F1 at every redshift bin and in a controlled bin-matched comparison. However, a substantial component of observed degradation is shared across all three models and reflects observational information loss rather than modeling capability: ellipticals collapse in recall at high redshift because their distinguishing surface-brightness features become invisible, and the lost mass is largely absorbed as false positives in the face-on non-spiral class—an over-prediction artifact visible in a precision-recall decomposition but hidden by raw accuracy. GBDT is uniquely brittle despite being higher-capacity than the SVM, which I attribute to the axis-aligned threshold structure of tree ensembles being a particularly poor match for smooth feature-distribution drift.

For astronomical pipelines trained at low redshift and applied to deeper surveys, the practical implication is that robust evaluation must include per-class metrics stratified by redshift, precision-and-recall decomposition rather than "accuracy," and ideally a controlled within-bin comparison to separate pure domain effects from redshift effects. Aggregate IID-vs-OOD macro-F1 alone is compatible with several very different failure-mode stories.

VI. GitHub

Code and data processing pipelines used in this study can be found at https://github.com/theonlynanu/GalaxyZooClassification.

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